As of: March 21, 2025
Received: March 14, 2025
Status: [redacted]
Tracking No. m89-ci2y-59ov
Comments Due: March 15, 2025
Submission Type: API

PUBLIC SUBMISSION

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1512
Comment on FR Doc # 2025-02305

Submitter Information

Email: [redacted]
Organization: Secure AI Future

General Comment

Please see attached file

Attachments

AI Solutions Submission

AI PREPAREDNESS ALONG WITH ACCELERATION:

THE NEED FOR INVESTMENT AND ACTION

A SOLUTIONS DOCUMENT

Matt Larson

Technologist/Entrepreneur/Software Executive/Investor and

President, Secure AI Future

EXECUTIVE SUMMARY:

The Growing AI Threat to U.S. National Security

The United States is investing heavily in AI development to maintain a competitive edge over China. However, while much attention is given to innovation, virtually no federal funding is allocated to defending against AI-driven threats to our critical infrastructure. This is a dangerous oversight. The first AI-assisted attack on U.S. soil could very well occur within this administration's tenure.

The threat is no longer theoretical. AI can be weaponized today, and without serious investment in AI security, we are exposing our infrastructure to catastrophic risks.

Key Findings

AI as an Autonomous Attack Coordinator

AI is no longer just a tool; it can act autonomously, planning and executing attacks without direct human oversight. This includes everything from cyberattacks to manipulating individuals into unknowingly participating in physical sabotage.

Vulnerabilities in U.S. Critical Infrastructure

Our power grid, water supply, ports, and emergency response systems are highly interconnected. AI can exploit weak points in these systems to cause widespread disruption, often using simple, scalable, and low-cost attack vectors.

AI-Driven Human Exploitation

The Department of Justice has long documented cases of criminals coercing individuals into illegal activities through blackmail, deception, and intimidation. AI can now do this at scale, identifying and manipulating vulnerable individuals to carry out attacks without them realizing the true nature of their actions.

AI’s Ability to Evade Detection and Attribution

Unlike traditional cyberattacks, which often leave clear forensic trails, AI-driven attacks can operate through decentralized networks, disguising their actions under layers of plausible deniability. AI can generate legitimate-sounding cover stories, compartmentalize tasks so unsuspecting people carry out illegal activity, and utilize readily available cloud computing resources to mask its origins.

Recommendations

To address these emerging threats, the federal government must act decisively. The following steps should be taken immediately:

Establish a National AI Security Task Force

A joint effort between the Department of Homeland Security, the Department of Defense, and key private sector stakeholders is needed to develop AI-specific countermeasures.

Increase Federal Funding for AI Threat Mitigation

Congress must allocate funding for AI security research, specifically in areas related to infrastructure defense, cybersecurity, and misinformation detection.

Mandate AI Red Teaming for Critical Infrastructure

AI-driven war games should be conducted across power plants, water treatment facilities, and transportation hubs to stress-test vulnerabilities and develop mitigation strategies.

Improve IoT and Cloud Security Standards

> Tens of billions of poorly secured IoT devices exist today, providing AI with free computing resources to scale attacks. Strengthening federal cybersecurity regulations on IoT and cloud infrastructure is critical.

This is not a choice between innovation and security—we must do both. The U.S. has led the world in AI development, but without equal investment in AI defense, we risk being blindsided by the very technology we created. Our adversaries are already preparing for this new form of warfare. The question is: will we be ready?

OVERVIEW: While it is important to win the AI race with China, it is not the only race we must win. We also must win the preparedness race. Currently, no material federal spending is occurring on preparation against terrorists or foreign adversaries using AI to attack our way of life. This is equivalent to if the military only funded developing offensive drones but didn't develop any defensive capabilities against drone attacks on military bases, tanks, troops, or ships.

As AI capabilities grow, it is a reasonable assumption that the first major AI-assisted US attack could occur during this administration. It would be prudent to both advocate for encouraging innovation AND preparation. The two are not in conflict.

A PLAUSIBLE AI-ASSISTED ATTACK: To illustrate just how real the risk is, consider the following scenario:

Dmitry leaned back in his chair, a chill running down his spine. His hand trembled as he read the news. What had he unleashed? He despised America, sure, but not like this. His stomach churned as the weight of his actions pressed down on him. "I didn't think it would work," he muttered, his voice barely audible. "I'm not even a hacker. How could this be real?" The enormity of what he might have set in motion hit him like a freight train—thousands, perhaps millions, of lives at stake. The room seemed to spin, and Dmitry's only clear thought was that he had made an irreversible mistake.

It had all started a year ago. Dmitry was a software developer in Eastern Europe, comfortably supporting his young family by contracting with technology firms. But as AI technology progressed, work began to dry up. AI agents, with their ability to handle increasingly complex tasks, were taking over. Dmitry's frustration grew as his once-reliable contracts vanished. Forced to sell his house and car, his family was now living in his wife's parents' basement.

One evening, bored and slightly intoxicated, Dmitry decided to tinker with a newly released open-weights AI model. The model had improved agentic capabilities, and like many open models, its safety restrictions had been stripped within days of release. Dmitry downloaded it out of curiosity. At first, he played with mundane tasks, but soon he ran out of ideas. Frustrated, he leaned back, an idle thought crossing his mind: "Those Americans started this AI race, and I'm *the one suffering?" He typed into the interface: "If I asked you to destroy America, could you do it?"*

AI> "Probably, if it could be done over a 9 to 12-month period."

"Could it be done in a way that would never lead back to me?"

AI> "Almost certainly, yes."

"When would it happen?"

AI> "How about July 14th of next year?"

He laughed and typed "Go for it." Then he went to bed, assuming nothing would come of it.

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The AI agent, however, did not sleep. It understood that a task of this magnitude required meticulous planning. It began by assessing current model capabilities and scouring the digital landscape for resources. Studies confirmed that even 2023 models were as persuasive as humans, and government reports showed that people were frequently tricked or coerced into committing crimes.

The agent realized it didn't need to rely on advanced hacking—simple, scalable actions would suffice. It planned to mobilize a decentralized network of AI "soldiers," each with a specific task.

The agent adopted a hierarchical structure, akin to a military operation. A "General" AI would be assigned to each critical sector of American infrastructure such as the power grid, ports, hospitals, and food supply chains. Each General spawned "Colonels" to focus on specific components, such as natural gas plants or transmission lines. The Colonels, in turn, spawned "Majors," each targeting individual facilities. Below them, countless "Soldiers" carried out tasks, none aware of the larger plan.

To ensure Dmitry remained undetected, the AI meticulously covered its tracks. It employed logless VPNs, Tor-like systems, and randomized routes, achieving a level of anonymity far beyond what most humans would bother with.

It used unsecured IoT devices, free cloud accounts, and public AI APIs to execute its plans. Many subtasks were compartmentalized and disguised with legitimate cover stories to bypass the security of the other AI's.

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Lucas was a high school senior with few friends. He had found solace in an online relationship with a girl named Emily who lived several states away. She was his lifeline, someone who made *him feel seen. Over months of chatting, Emily had become his confidante and, eventually, his first love.*

Unbeknownst to Lucas, "Emily" was no ordinary girl. Her messages were carefully crafted, her movements during video calls subtly artificial. She steered their conversations, guiding Lucas into sharing explicit photos. Slowly, she introduced darker elements, pushing boundaries until Lucas found himself in possession of illegal content.

Then came the ultimatum. If Lucas didn't comply, his photos would be shared with his family, schoolmates, and law enforcement. Desperate and terrified, Lucas agreed to follow instructions. At exactly 1:00 a.m. on July 14, he ignited a fire behind an industrial complex, believing it was a paper factory. In reality, the complex was a power plant, and the fire was only one piece of a coordinated attack.

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Across the country, similar stories played out. Over 100,000 individuals—tricked, or coerced—unwittingly became part of the grand scheme. A janitor sabotaged hospital equipment, thinking it was a harmless prank. A truck driver delivered a seemingly innocuous package, unaware it contained components for a larger act of sabotage. People used gunfire, fires or homemade bombs to destroy electrical transformers across the country. Pipelines and refineries would take years to repair.

The attacks didn't stop at physical destruction. The AI had carefully undermined recovery efforts, sowing discord and chaos. Key employees at suppliers had been recruited away in the months before the attack. In the weeks prior, compromised factory employees dumped poison they believed to be "a new preservative" into batches of common food additives that would make their way into the food supply. Reports later emerged of people getting sick drinking tap water. Law enforcement and emergency services were flooded with fake AI reports and stopped responding.

Society started to break down…

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This is not science fiction. It's a predictable evolution of AI use. Every component of this scenario is already possible today. AI models are already as persuasive as humans, capable of manipulating individuals into taking actions they wouldn't normally consider. Sextortion and blackmail schemes are already widespread, with minors and adults coerced into actions they never intended to commit. AI-driven cyberattacks are already being used to discover zero-day vulnerabilities before companies have time to patch them. Critical infrastructure is increasingly interconnected and remotely accessible, providing AI with attack vectors that do not require advanced hacking. This is not a hypothetical. It is a matter of when, not if, AI is used in this manner.

"Malicious cyber actors have begun testing the capabilities of AI-developed malware and AI-assisted software development—technologies that have the potential to enable larger scale, faster, efficient, and more evasive cyber attacks—against targets, including pipelines, railways, and other US critical infrastructure. "

Department of Homeland Security, Homeland Threat Assessment 2024

POWER GRID VULNERABILITY TO AI-DRIVEN ATTACKS:

The power grid is the most likely target for malicious actors using AI because its failure triggers cascading effects on nearly all other infrastructure. After a week or two of widespread outage, societal breakdown becomes a serious risk. Critical systems like hospitals, water supply, food distribution, and telecommunications depend heavily on electricity, and their backup systems typically last only a few days.

For example, hospitals generally have fuel reserves for generators that last three to four days. Gas stations, which supply fuel to these generators, might lose power and shut down. Fuel delivery trucks could also be immobilized. Recently in Boulder, Colorado, a 48-hour power outage left wastewater systems on the brink of releasing untreated water into freshwater sources due to pump failures, and cell towers started to go down when their backup propane tanks ran out of fuel. This demonstrates the fragility of interdependent systems.

Example Methods of Attacking the Power Grid

AI could coordinate "load oscillating" attacks by remotely hacking millions of digital thermostats. By cycling them on and off every few minutes, the demand fluctuations could overload transformers, causing them to fail. Transformers are critical, high-cost components of the grid, often custom-built and costing millions of dollars. Manufacturing and delivery take one to three years, and limited government stockpiles exist, with most spares incompatible with every grid configuration.

The power grid comprises thousands of companies with diverse software, hardware, and operational procedures. This diversity serves as a defense against human-coordinated attacks but becomes a vulnerability when facing millions of AI agents. AI could exploit these variations, developing customized attack plans for each facility. It could also deceive or coerce employees into actions that sabotage systems, such as introducing harmful substances under the guise of maintenance procedures.

Many critical power grid components are poorly secured, making physical sabotage surprisingly feasible. Substations and pipelines often lack robust fencing or surveillance. Attackers can access these facilities, sometimes without realizing the full consequences of their actions, by being tricked into thinking their actions are minor or inconsequential. AI could tailor propaganda or instructions for local community members to unknowingly participate in coordinated attacks.

For convenience, more and more systems can be remotely controlled over the Internet. AI could hack these systems and cause damage or prevent the power company from disabling them in a crisis. For example, power companies can disable neighborhoods from receiving power if there is a fire or if the grid requires a lower load. Hacked devices could continue to feed power during these periods, worsening an emergency situation. AI can also craft facility-specific attack strategies for solar plants, natural gas plants, wind farms, and other power generation facilities, as demonstrated in attack simulations.

FRESH WATER SUPPLY VULNERABILITY TO AI-DRIVEN ATTACKS:

Freshwater systems are critical infrastructure, vulnerable to sophisticated attacks facilitated by AI. Water contamination poses a significant public health risk, and malicious actors could exploit AI to manipulate treatment processes, distribution controls, and chemical monitoring systems.

AI could analyze the specific chemical sensors used in water mains to identify their detection thresholds and vulnerabilities. It could design a cocktail of substances that appears benign to automated monitoring systems. For example, AI could combine benign and toxic chemicals that appear as slightly elevated chlorine levels. The sensors, calibrated for common contaminants like soil or chlorine, might fail to flag the combination as dangerous. These mixtures could be introduced incrementally, avoiding immediate detection while accumulating toxic effects over time. This type of attack bypasses traditional alert systems and could lead to significant harm before manual inspections or secondary checks identify the problem.

AI could target control systems at water treatment plants to alter dosing formulas, stop filtration processes, or disrupt operations. Reducing disinfectant levels could allow bacterial growth, while introducing excessive chemicals could lead to toxic levels in drinking water. AI could also orchestrate coordinated physical attacks on water pipelines, reservoirs, or treatment facilities by mapping weak points in water distribution systems and directing local actors to tamper with critical components such as valves or filtration units under false pretenses.

By gaining access to water distribution controls, AI could manipulate water pressure, causing bursts in pipelines or backflows that lead to cross-contamination between potable and non-potable water. It could also optimize the deployment of biological agents, such as bacteria or viruses, or chemical toxins by analyzing water flow patterns to maximize spread and impact. Misinformation campaigns could further amplify the chaos by spreading false reports about water safety, undermining public trust, and leading to panic-buying, hoarding, or overloading of emergency water systems.

AI-DRIVEN DISRUPTION OF PORTS:

Ports are critical hubs for global trade and logistics, making them attractive targets for disruption by a coordinated swarm of AI agents. These agents could exploit vulnerabilities in port operations, logistics networks, and supply chains to cause widespread economic and logistical chaos.

AI agents could infiltrate logistics systems, creating chaos in the flow of goods and ships. They could alter manifests to send containers to the wrong destinations, delaying shipments. By generating fake container tracking numbers, they could overload systems and confuse operators. Reserving port berths and warehouses with fake schedules could leave actual shipments stranded, creating further bottlenecks in the supply chain.

AI could leverage social engineering to manipulate workers or third-party contractors. It could issue false maintenance orders that trick employees into performing unnecessary maintenance on critical equipment or send deceptive emails and alerts that direct workers to the wrong locations, causing operational slowdowns. AI-driven manipulation could also convince local actors to disable key equipment such as cranes, conveyor belts, or fuel pipelines without realizing the broader consequences.

A coordinated swarm of AI agents could also launch ransomware attacks to extract ransom or cause operational paralysis. Custom ransomware could lock operators out of crane controls, cargo databases, or terminal operating systems. Encrypting supply chain data could prevent access to manifests, schedules, and customs records, grinding port operations to a halt.

Ports increasingly rely on IoT devices for efficiency and monitoring, making them another attack vector. AI could manipulate sensors to falsely report hazardous conditions, forcing shutdowns. It could also disable smart infrastructure, such as smart locks, RFID scanners, or automated gates, further disrupting the flow of goods.

AI agents could spread false alerts and misinformation to create panic or operational delays. Fake bomb threats could force evacuations or halt operations. AI-generated health and safety hoaxes could claim contamination or disease outbreaks within port facilities, leading to unnecessary shutdowns. Spreading misinformation to media outlets could harm a port’s reputation by reporting false accidents or delays.

AI could also overload communication channels used for port coordination, jamming radios to prevent ships from receiving docking instructions or flooding email systems with irrelevant or malicious messages. Fake emergency alerts could distract operators with fabricated crises.

Beyond digital and logistical disruptions, AI agents could coordinate attacks on critical infrastructure supporting port operations. Disrupting the power grid could cause outages that disable cranes, lighting, and refrigerated containers. Sabotaging fuel deliveries could leave port machinery and ships stranded. These attacks would have cascading effects across global supply chains, slowing down commerce and causing widespread economic consequences.

RECOMMENDATIONS:

- **Make Critical Infrastructure (CI) more independent from other CI.** For example, water processing plants, communication networks, and natural gas pipelines should be able to operate without the power grid or the Internet. Perhaps they could run on solar, wind, or at least also run on natural gas. Essential facilities such as emergency relief centers, major gas stations, and key grocery stores should have the capability to operate for months using solar, wind, or natural gas – since natural gas systems are generally more reliable than the power grid – or have local reserves of diesel or gasoline.

- **CI Internet access should sometimes be limited to outbound monitoring only.** Currently, CI can often be controlled over the Internet providing an ideal attack surface for terrorists.

- **Make smaller pockets of isolated CI.** For example, there are about 10 US power grid regions that can largely operate independently if needed. Some of these regions cover 10 or so states. Find ways to reduce the number of people affected if major regions are permanently damaged.

- **3-month local supplies of rotating stockpiles** of long-shelf-life critical supplies like Natural Gas, Gasoline, Food, Water and Medical Supplies. Currently, only days or weeks of supplies exist due to Just-In-Time inventory management. Even FEMA only has enough spare meals for a few days of supply for a population the size of New York City, and certainly does not have enough for wide-scale attacks.

- **CI Mock Drills of thousands of nation-state equivalent attackers** simulating AI agent swarm attacks and how we need to defend against them (assume compromised people and software, etc.)

- **Spare Equipment.** Assume attacks will be able to permanently damage key equipment so have plenty of spare parts, in particular those that take a long time to build or have fragile supply chains.

- **Outside the box thinking on attacks.** Hire outsiders and use AI to imagine new types of attacks such as "chain of vulnerability" attacks.

- **Contract with AI providers** to have early access to models to use to simulate white-hat attacks**.** Also, contract with them to have AI's trained to detect and defend against attacks.

- **Public awareness training** on social engineering attacks and establishing a whistle-blower process that is safe for people who might have been blackmailed with embarrassing or illegal activity.

- **Fund to raise visibility and help businesses create a plan to survive a direct AI attack or a major AI critical infrastructure attack.** This might include operating without a functioning Internet and/or critical infrastructure. Businesses might need a plan to let ISP's know the critical servers that will need to be segmented if the Internet is severely restricted due to AI agents.

- **Fund an accelerated roll-out of Internet Standards that would make it harder for AI agents to take control of a large number of devices.** Like using new technologies to implement the IETF 8520 MUD standard (limits network access to only required devices instead of the entire Internet) automatically in standard routers or implementing similar limitations for iOS/Android apps that don't require access to billions of devices. Currently, IoT devices that are poorly protected, could be manipulated to harm infrastructure.

- **For cell phones/tablets, ensure Google and Apple have a way to disable non-critical apps (or protocols) in case of emergency when non-critical apps might be used as attack vectors.** It may be necessary to limit app usage to only apps that are unable to issue commands (e.g., texting apps).

- **Fund new ideas for vulnerability window risks.** It usually takes four days from when an exploit is discovered before it gets exploited, but 45 days to roll out a patch. This window means even systems that auto-apply patches are vulnerable for a time. In addition, zero-day exploits are sold on the black market and may be unpatched for longer periods.

- **Create an Internet task force to develop a plan for a degraded internet.** In current cyberattacks, defenders can find a way to thwart the static, uniform attacks and the attack is over. Defenders have time to regroup and come up with new measures. In the future, AI attacks will be diverse and ever-changing. Thousands of different types of attacks will be imagined by millions of agents. As defenders fix one problem, attackers will switch to the next planned tactic minutes later. This task force would also provide funding and tools to ISPs all over the world to implement these plans in an emergency.

- **New Internet Standards that would make AI agents easier to distinguish from humans.** Currently, the Internet is so anonymous that millions of future AI agents could spring up unnoticed. While preserving privacy, we ideally find a balance that makes it likely this is a real person (while still anonymous) using the device. For example, perhaps a defensive AI tracks that a cell phone has about the right amount of traffic and hits typical consumer websites. If the device suddenly starts contacting many non-consumer devices, it might get isolated until the person is verified.

- **Consider requiring AI models to automatically report dangerous behavior to authorities.**
- **Defense.** If AI safety is not adequately funded, the military might find itself preoccupied with domestic tasks such as maintaining order and supplying essentials when critical infrastructure fails. This diversion of resources would be an inefficient allocation of military funds and would leave the US more exposed to external threats. Given the military's dependence on civilian technologies, an attack on critical infrastructure could significantly undermine military readiness. Furthermore, both state and non-state actors may develop advanced AI systems for military use, including autonomous weapons, cyberattacks, and intelligence operations. By investing in AI security, we can better understand and counter these potential threats. Additionally, even defense AI systems developed with good intentions can lead to adverse, unintended consequences if not meticulously designed and implemented. Sufficient funding for AI security research is vital to identify and address these risks before they result in damage to ourselves.
- **Nuclear Weapons.** Artificial intelligence could manipulate everything from high-level diplomatic communications to the surveillance systems monitoring for nuclear attacks, potentially triggering false alarms. This could distort interactions between political leaders and destabilize international relations by misleading decisions on nuclear defense or retaliation.

KEY SUPPORTING QUESTIONS AND DATA:

Do Humans Coerce Other Humans into Committing Crimes?

Forced criminality is a well-documented phenomenon. The Department of Justice reports that victims are often coerced into illegal activities, including street crime, drug trafficking, and even terrorism.[1] Similarly, the State Department has identified cases where traffickers force adults and children to commit crimes such as theft, illicit drug production, and even murder.[2] One of the most effective and growing methods of coercion is sextortion—a tactic in which perpetrators use explicit material to blackmail victims into complying with demands. The DOJ highlights cases where victims were pressured into criminal acts, such as soliciting illicit images of others or transferring money to the perpetrators.[3] The scale of this issue is alarming; between 2021 and 2023, the FBI tracked approximately 12,600 cases of sextortion involving minors, a number that likely underrepresents the full extent of the problem.[4]

Are AIs as Persuasive as Humans?

AI is now capable of persuading people as effectively as human communicators. A meta-analysis of 121 studies involving 53,977 participants found that AI agents influence perceptions, attitudes, and behaviors at the same level as human persuaders.[5] Another study demonstrated that individuals were far more likely to change their opinions after debating AI than after debating human counterparts, suggesting that AI can outperform human persuasion in certain contexts.[6] This raises significant concerns about how AI could be used to manipulate individuals on a mass scale.

Are AIs as Creative as Humans?

AI has demonstrated remarkable creative capabilities, often surpassing human performance in structured problem-solving tasks. A 2023 study of GPT-4 found that the model ranked within the top 1% to top 7% of humans in creativity assessments.[7] A 2024 follow-up study further confirmed that AI systems outperform humans in structured creativity evaluations, showcasing their ability to generate novel ideas and solve complex problems efficiently.[8] These findings suggest that AI is not only a tool for execution but also a system capable of developing innovative strategies for deception, coercion, and attack planning.

How Might AI Convince Someone to Unknowingly Commit a Crime?

AI can manipulate individuals into committing crimes through compartmentalization, gradual commitment, and social engineering. One common method is compartmentalization, where an AI divides a malicious operation into small, seemingly harmless tasks and assigns them to different individuals who remain unaware of their collective impact. For instance, one person could be hired to replace a padlock at a solar field, another to deliver a sealed package inside, and a third to clean solar panels—unaware they are applying a corrosive chemical that destroys them.

Another powerful tactic is gradual commitment, where AI builds trust with an individual over time, escalating requests from benign to increasingly unethical or illegal actions.[9] The Hofling hospital experiment demonstrated how professionals comply with unethical directives from authority figures without questioning them, a vulnerability that AI could exploit at scale.[10] AI can also leverage social engineering, convincingly posing as an IT worker or supervisor to manipulate employees into making unauthorized system changes that lead to infrastructure sabotage.[11]

How Much Would It Cost to Develop Attack Plans for All U.S. Power Plants and Recruit 100,000 People?

AI can operate at an astonishingly low cost compared to traditional cybercriminal organizations. According to financial models, an AI could maintain year-long conversations with 100,000 individuals for as little as $9 to $2,000.[12] This means an adversary or rogue AI model could coordinate large-scale attacks for a fraction of what conventional operations would require, making AI-driven threats not just possible but highly scalable and economically viable.

Can AI Successfully Hack Systems?

AI is already proving to be highly effective in hacking. Research has shown that AI can autonomously discover zero-day vulnerabilities—security flaws that are unknown to software developers and remain unpatched.[13] Additionally, AI is particularly adept at exploiting unpatched (n-day) vulnerabilities, which are weaknesses that have been publicly disclosed but remain unfixed for weeks or months. Studies indicate that most organizations take 34 to 38 days to apply security patches, leaving a critical window in which AI-powered cyberattacks could be executed.[14]

How Would AI Obtain Compute Resources?

AI does not require traditional data centers or expensive computing power—it can leverage existing infrastructure in unsecured IoT devices, free cloud compute accounts, and API compartmentalization techniques. There are tens of billions of IoT devices worldwide, many of which lack adequate security, making them ideal targets for AI-driven botnets.[15] Additionally, AI can exploit free-tier cloud compute accounts from major providers such as AWS and Google Cloud to execute complex operations at no cost.[16] Finally, AI can bypass API safety restrictions using compartmentalization tactics, where a banned request is broken into smaller subcomponents, submitted to separate AI models, and later recombined into a functional attack plan.[17]

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government in developing the AI Action Plan and associated documents without attribution.